

Thermal convection

Fluid motion driven by thermal gradients, also called THERMAL CONVECTION, is a familiar and important process in nature.

Beyond its numerous technological applications, it is the basis for the interpretation of several phenomena as:

- the drift of the continental plates,
- the sun activity,
- the large-scale circulations observed in the oceans,
- the atmosphere, ...

As a prototype of thermal convection, it will be examined the behaviour of a thin fluid layer enclosed between two horizontal surfaces whose lateral dimensions are much larger than the width of the layer.

The two horizontal bounding planes are either rigid or stress-free surfaces,

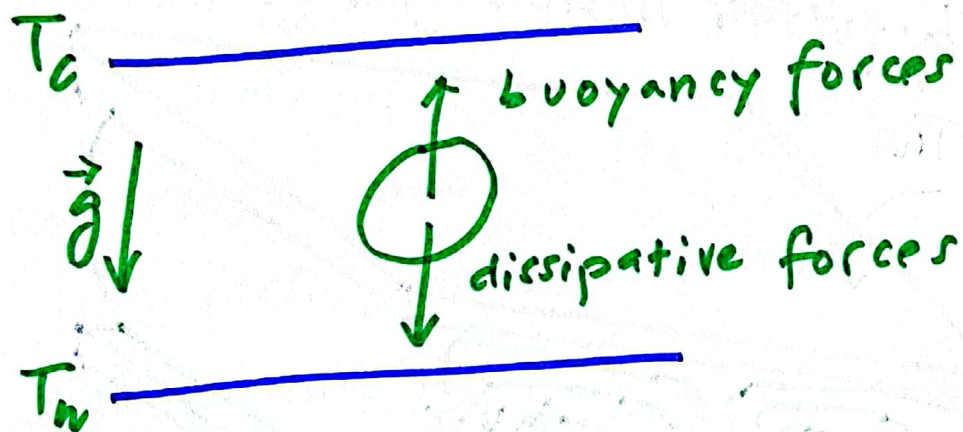
the lower surface is uniformly heated so that the fluid is subject to a vertical temperature gradient.

The most influential experimental investigation on thermal convection dates back to Bénard (1900). The fluid used by Bénard was molten spermaceti, a whale's non-volatile viscous oil, and the motion was made visible by graphite or aluminium powder. In Bénard's original experiment, the lower surface was a rigid plate but the upper one was open to air, which introduces an asymmetry in the boundary conditions besides surface tension effects.

The essential result of Bénard's experiment was the occurrence of a stable, regular pattern of hexagonal convection cells.

A first theoretical interpretation of thermal convection was provided by Rayleigh (1920)

We are going to study the instabilities occurring in a viscous fluid layer of thickness d (between a few millimetres and a few centimetres) and infinite horizontal extent limited by two horizontal non-deformable free surfaces, the z -axis pointing in the opposite direction of the gravity acceleration $\vec{g}$



The fluid is heated from below with T_h and T_c , the temperatures of the lower and upper surfaces, respectively.

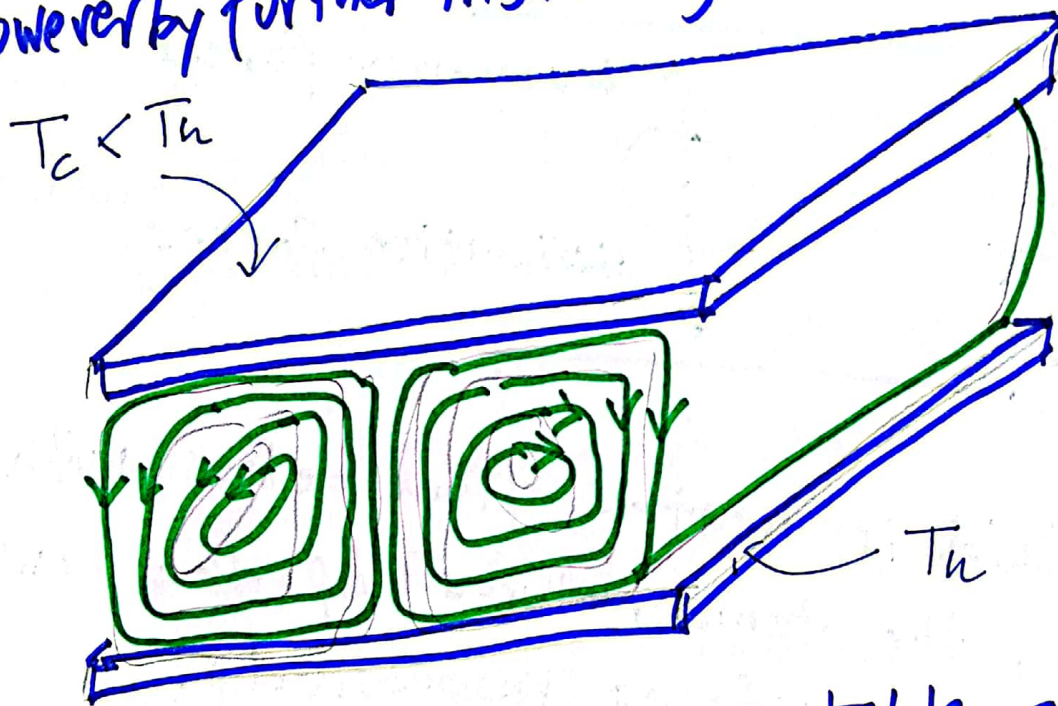
The mass density ρ is assumed to be

$$\rho = \rho_0 [1 - \alpha (T - T_0)]$$

where T_0 is an arbitrary reference temperature, and α the coefficient of thermal expansion, generally a positive quantity (α is of the order of $10^{-3} - 10^{-4} \text{ K}^{-1}$)

When the temperature difference $\Delta T = T_h - T_c$ (typically not more than a few $^{\circ}\text{C}$) between the two bounding surfaces is lower than some critical value, no motion is observed and heat propagates only by conduction inside the fluid.

However by further increasing ΔT , the basic heat



conductive state becomes unstable at a critical value $(\Delta T)_{\text{crit}}$ and matter begins to perform bulk motions which, in rectangular containers, take the form of regular rolls aligned parallel to the short side as visualized in the figure, this structure is referred as a roll pattern. Note that the direction of rotation of the cells is

unpredictable and uncontrollable, and that two adjacent rolls are rotating in opposite directions.

Qualitative interpretation of the onset of motion.

By submitting the fluid layer to a temperature difference, one generates a temperature and a density gradient.

A fluid droplet close to the hot lower plate has a lower density than everywhere in the layer, as density is generally a decreasing function of temperature. As long as it remains in place, the fluid parcel is surrounded by particles of the same density, and all the forces acting on it are balanced.

Assume now that, due to a local fluctuation, the droplet is slightly displaced upward. Being surrounded by cooler and denser fluid, it will experience a net upward Archimede's buoyant force proportional to its volume and temperature difference whose effect is to

amplify the ascending motion.

Similarly, a small droplet initially close to the upper cold plate and moving downward will enter a region of lower density and becomes heavier than the surrounding particles. It will therefore continue to sink, amplifying the initial descent.

What is observed in the experiments is thus the result of these upward and downward motions.

However, experience tells us that convection does not appear whatever the temperature gradient as could be inferred from the above argument. The reason is that stabilizing effects oppose the destabilizing role of the buoyancy force;

one of them is VISCOSITY, which generates a friction force directed opposite

to the motion, the second one is HEAT DIFFUSION, which tends to spread out the heat contained in the droplet towards its environment reducing the temperature difference between the droplet and its surroundings.

This explains why a critical temperature difference is necessary to generate a convective flow: MOTION WILL START AS SOON AS BUOYANCY OVERCOMES THE DISSIPATIVE EFFECTS OF VISCOUS FRICTION AND HEAT DIFFUSION.

These effects are best quantified by the introduction of the thermal diffusion time

$$\tau_{\chi} = \frac{d^2}{\chi} \leftarrow \begin{array}{l} \text{thermal} \\ \text{diffusivity} \end{array}$$

and the viscous relaxation time

$$\tau_{\nu} = \frac{d^2}{\nu} \leftarrow \begin{array}{l} \text{Kinematic} \\ \text{Viscosity} \end{array}$$

τ_x is the time required by the fluid to reach thermal equilibrium with its environment.

τ_v is related to the time needed to obtain mechanical equilibrium.

Another relevant timescale is the buoyancy time, i.e., the time that a droplet, differing from its environment by a density defect $\delta\rho = \rho_0 \alpha \Delta T$ needs to travel across a layer of thickness d .

$$\tau_B^2 = \frac{d}{\alpha g \Delta T}$$

This result is readily derived from Newton's law of motion $\rho_0 \frac{d^2 z}{dt^2} = g \delta\rho$ for a small volume element. A large value of ΔT means that the buoyancy time is short.

To give an order of magnitude of these various timescales, let us consider a shallow layer of silicone oil characterized by

$$d = 10^{-3} \text{ m}$$

$$\nu = 10^{-4} \text{ m}^2 \text{ s}^{-1}$$

$$\chi = 10^{-7} \text{ m}^2 \text{ s}^{-1}$$

it is then found that

$$\tau_\chi = 10 \text{ s}$$

$$\tau_\nu = 10^{-2} \text{ s}$$

The relative importance of the buoyant and dissipative forces is obtained by considering the ratios τ_ν/τ_B and τ_χ/τ_B or, since they occur simultaneously, through the so-called dimensionless Rayleigh number

$$Ra = \frac{\tau_v \tau_x}{\tau_b^2} = \frac{\alpha g \Delta T d^3}{\nu \chi}$$

The Raileigh number can be viewed as the ratio between the destabilizing buoyancy force and the stabilizing effects expressed by the viscous drag and the thermal diffusion; convection will start when Raileigh number exceeds some critical value $(Ra)_c$.

- For $Ra < (Ra)_c$, the fluid remains at rest and heat is only transferred by conduction,
- for $Ra > (Ra)_c$, there is a sudden transition to a complex behaviour characterized by the emergence of order in the system.

The ratio between the dissipative processes is measured by the dimensionless Prandtl number defined as

$$Pr = \frac{\tau_x}{\tau_v} = \frac{\nu}{\chi}$$

For gases $Pr \sim 1$, for water $Pr \sim 7$,
for silicone oils $Pr \sim 10^3$,
for Earth's mantle $Pr \sim 10^{23}$

In a linear stability approach, the main problem is the determination of the marginal stability curve, i.e., the curve of Ra vs. the wave number k at $\lambda = 0$.

Thermal Convection. Rayleigh-Bénard.

⊗ Equations of fluid mechanics written within the Boussinesq approximation:

- density is considered to be constant except the buoyancy term
- viscosity, thermal diffusivity and thermal expansion coefficient are temperature independent
- mechanical dissipated energy is negligible

⊗ The governing equations of mass, momentum and energy balance are then given by

$$\nabla \cdot \vec{v} = 0$$

kinematic
viscosity

$$\left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v} \right) = - \frac{1}{\rho_0} \nabla p + [1 - \alpha(T - T_0)] \vec{g} + \nu \nabla^2 \vec{v}$$

$$\frac{\partial T}{\partial t} + \vec{v} \cdot \nabla T = \chi \nabla^2 T \quad (1)$$

thermal diffusivity $\chi = \frac{k}{\rho c_p}$

The set represents five scalar partial differential equations for the five unknowns p, T, v_x, v_y, v_z

$$\rho = \rho_0 [1 - \alpha(T - T_0)]$$

T_0 : arbitrary reference temperature

coefficient of thermal

expansion, generally a positive quantity except for water around 4°C . For ordinary liquids, $\alpha \sim 10^{-3} - 10^{-4} \text{ K}^{-1}$

⊗ In the basic unperturbed state, the fluid is at rest and temperature is conveyed by conduction, so that the solutions of (1) are simply

$$V_r = 0, \quad T_r = T_h - \frac{\Delta T}{d} z, \quad \frac{\partial p_r}{\partial z} = -\rho_0 g [1 - \alpha(T_r - T_0)] \quad (2)$$

Where subscript r refers to the unperturbed reference solution, z denotes vertical coordinate measured positive with $z=0$ corresponding to the lower boundary, and x, y the horizontal coordinates, and $\Delta T = T_h - T_c$ is the positive temperature difference between the lower and the upper boundaries.

⊗ Designating by

$$\begin{aligned} \vec{v}' &= \vec{v} - \vec{0} \\ T' &= T - T_r \\ p' &= p - p_r \end{aligned}$$

the infinitesimally small perturbations of the basic state, we can linearize (1) and obtain the following set for the perturbed fields

$$\nabla \cdot \vec{v}' = 0 \quad (3)$$

$$\frac{\partial \vec{v}'}{\partial t} = -\frac{1}{\rho_0} \nabla p' + \alpha T' g \hat{e}_z + \nu \nabla^2 \vec{v}' \quad (4)$$

Unit vector pointing opposite to $\vec{g}$

$$\frac{\partial T'}{\partial t} + \vec{v}' \cdot \nabla T_r = \chi \nabla^2 T' \quad (5)$$

⊗ Boundary conditions.

- Perfectly heat conducting surfaces, $T' = 0$

We assume that the thermal conductivity at the limiting surfaces is much higher than in the fluid itself, any thermal disturbance advected by the fluid will be instantaneously smoothed out so that T' will be vanish at the boundary surfaces.

- Adiabatically isolated surfaces, $\partial T' / \partial z = 0$
The surfaces are impermeable to heat flow.

- Heat transfer is governed by Newton's cooling law
$$-\lambda \partial T / \partial z = h(T - T_\infty)$$

- Free surfaces, the shearing stress is zero at the surface, when use is made of equations of continuity (3), this condition is identical to setting $\partial^2 v'_z / \partial z^2 = 0$ together with $v'_z = 0$, as the surfaces are non-deformable.

- Rigid boundaries, the non-slip condition imposes that all the components of velocity are zero $v'_x = v'_y = v'_z = 0$, which combined with (3) yields
$$v'_z = \partial v'_z / \partial z = 0$$

⊗ A further simplification of the set (1)-(3) is obtained by applying twice the operator rot ($\equiv \nabla \times$) on the momentum equation and using the continuity equation, we are then left with the two following relations in the two unknowns v'_z and T'

$$\frac{\partial}{\partial t} (\nabla^2 v'_z) = \alpha g \nabla_1^2 T' + \nu \nabla^4 v'_z \quad (8)$$

$$\frac{\partial T'}{\partial t} = \beta v'_z + \chi \nabla^2 T' \quad (9)$$

where $\nabla_1^2 \equiv \partial^2 / \partial x^2 + \partial^2 / \partial y^2$ denotes the horizontal Laplacian and $\beta \equiv \Delta T / d$.

⊗ We now make the variables dimensionless by introducing the following scaling

$$X = \frac{x}{d} \quad Y = \frac{y}{d} \quad Z = \frac{z}{d}$$

$$\hat{t} = \frac{\nu}{d^2} t \quad w = \frac{d}{\nu} v'_z \quad \theta = \frac{\chi}{\nu} \frac{T'}{\Delta T} \quad (10)$$

⊗ Normal mode solution of the form

$$W = W(z) \exp[i(k_x X + k_y Y)] \exp(\lambda \hat{t}) \quad (11)$$

$$\theta = \theta(z) \exp[i(k_x X + k_y Y)] \exp(\lambda \hat{t}) \quad (12)$$

Where $W(z)$ and $\theta(z)$ are the amplitudes of the perturbations, k_x and k_y are the dimensionless wave numbers in the directions x and y , respectively, and λ is the dimensionless growth rate.

⊗ Since the present problem is self-adjoint, it can be proved that λ is a real quantity.

⊗ Substitution of solutions (11) and (12) in (8) and (9) leads to the following amplitude differential equations

$$\left(\frac{d^2}{dz^2} - k^2\right) \left(\frac{d^2}{dz^2} - k^2 - \lambda\right) W = Ra k^2 \theta \quad (13)$$

$$\left(\frac{d^2}{dz^2} - k^2 - \lambda Pr\right) \theta = -W \quad (14)$$

where $k^2 = k_x^2 + k_y^2$

$$Ra = \frac{\tau_v \tau_x}{\tau_b^2} = \frac{\alpha g \Delta T d^3}{\nu \chi}$$

Rayleigh number

$$Pr = \tau_x / \tau_v = \nu / \chi$$

Prandtl number

⊗ The boundary conditions corresponding to two perfectly heat conducting free surfaces are

$$W = \frac{d^2 W}{dz^2} = 0, \quad \theta = 0 \text{ at } z=0 \text{ and } z=1 \quad (15)$$

The latter are satisfied for solutions of the form

$W = A \sin \pi z$, $\theta = B \sin \pi z$, which substituted in (13) and (14) lead to the following algebraic equations for the two unknowns A and B :

$$A + (\pi^2 + k^2 + \lambda) B = 0 \quad (16)$$

$$(\pi^2 + k^2)(\pi^2 + k^2 + \lambda Pr^{-1}) A - k^2 Ra B = 0$$

Non-trivial solutions demand that the determinant of the coefficients vanishes, which results in the following dispersion relation between k , λ , Ra and Pr

$$(\pi^2 + k^2) \lambda^2 + (\pi^2 + k^2)^2 (1 + Pr) \lambda + \dots$$

$$\dots + Pr [(\pi^2 + k^2)^3 - k^2 Ra] = 0 \quad (17)$$

⊗ By setting $\lambda = 0$ in (17), one obtains the marginal curve $(Ra)_0$ vs. k determining the Rayleigh number at the onset of convection (see Figure 1); it is independent of the Prandtl number and given by

$$(Ra)_0 = \frac{(\pi^2 + k^2)^3}{k^2} \quad (23)$$

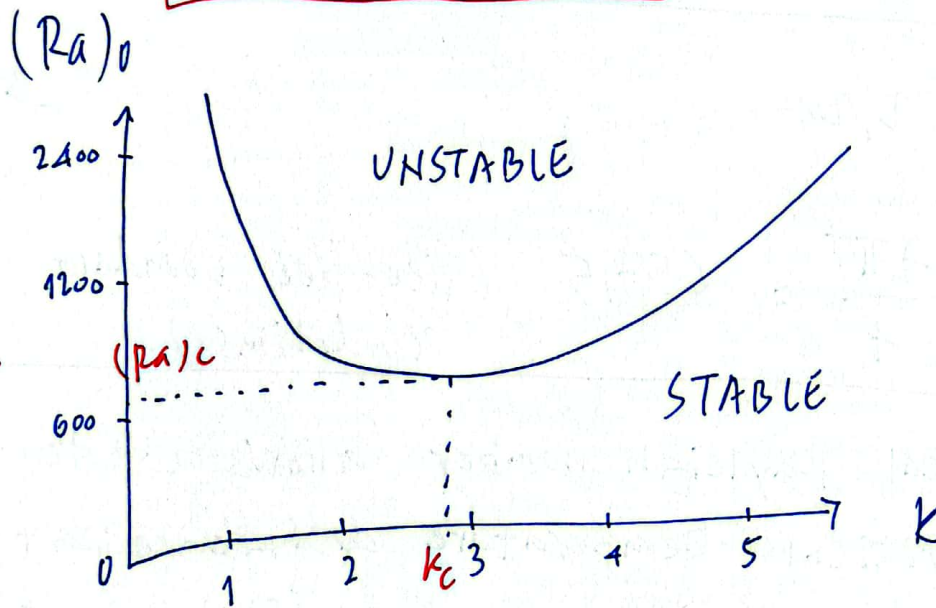


Figure 1

Marginal stability curve for Rayleigh-Bénard's instability in a horizontal fluid layer limited by two stress free perfectly heat conducting surfaces.

⊗ The solutions of (17) can be written as :

$$\lambda = \frac{(\pi^2 + k^2)(1 + Pr)}{2} \left\{ -1 \pm \left[1 + \frac{4[Ra - (Ra)_0]}{(Ra)_0(1 + Pr)(1 + Pr^{-1})} \right]^{1/2} \right\} \quad (24)$$

⊗ For $Ra < (Ra)_0$, one has $\lambda < 0$ (stability) while for $Ra > (Ra)_0$, $\lambda > 0$ (instability)

⊗ λ is real when $Ra > 0$ and that $\lambda < 0$ when $Ra < 0$, this latter result indicates that the system is unconditionally stable by heating from above.

⊗ The minimum value of the marginal curve is obtained from (23) by differentiation with respect to k ; setting this result to zero gives the critical wave number k_c at which the curve $(Ra)_0(k)$ is minimum, and the corresponding critical Rayleigh number $(Ra)_c$,

$$k_c = \frac{\pi}{\sqrt{2}} = 2,21$$

(26)

$$(Ra)_c = \frac{27\pi^4}{4} = 657,5 \quad (\text{free-free boundary conditions})$$

⊗ The critical Rayleigh number allows us to determine the critical temperature difference at which the system changes from the state of rest to the state of cellular motion, the critical wave number provides information about the horizontal periodicity of the patterns at the onset of convection, k_c represents the most dangerous mode picked up by the fluid.

⊗ For other boundary conditions, the calculations are more complicated. but the procedure remains valid, the marginal curves will have approximately the same form as in Figure 1 with critical values given by

$k_c = 2,68$	$(Ra)_c = 1100,6$	(rigid-free boundaries)
$k_c = 3,117$	$(Ra)_c = 1707,7$	(rigid-rigid boundaries)

⊗ As expected, stability is reinforced (larger $(Ra)_c$ value) in presence of rigid surfaces as the fluid motion is more strongly inhibited by the viscous forces. At the same time the dimensions of the cells (larger k_c value) are diminished: more energy must now be dissipated to compensate for the larger release of energy by buoyancy, clearly narrower cells are associated with greater dissipation and energy release.